

Arjuna JEE 2.0 2027

Maths

Basic Mathematics

DPP: 11

Total Duration - 25 Min

Q1 If $\log_{10} 2 = 0.3010$, $\log_{10} 3 = 0.4771$. Find the number of integers in :

- a) 5^{200}
 b) 6^{15}
 c) The number of zeros after the decimal in 3^{-100} .

Q2 Solve: $1 + 2 \log_{x+2} 5 = \log_5(x+2)$

Q3 Solve: $\log_{1-x}(3-x) = \log_{3-x}(1-x)$

Q4 Solve: $7^{\log x} = 98 - x^{\log 7}$

Q5 If $\log 2 = 0.3010$ then number of integers in the expansion of 4^{17} is

- (A) 9 (B) 11
 (C) 13 (D) 15

Q6 If $\log_5(\log_5(\log_2 x)) = 0$, then the value of x is

- (A) 32 (B) 125
 (C) 625 (D) 126

Q7 Integral value of x which satisfies the equation $\log_6 54 + \log_x 16 = \log_{\sqrt{2}} x - \log_{36}(4/9)$ is _____

Q8 If the value of the expression

$$E = 5^{\log_{\sqrt{3}} 3} + \log_{\sqrt{3}} \left(\frac{9}{\sqrt{2}+1} \right) + \log_{1/3} \text{ is}$$

$$\left(\frac{1}{3+\sqrt{8}} \right)$$

equal to M , then the value of $M - 6$ is equal to

Q9 Solve for x : $7^{x+8} = 8^{x+7}$

Q10 Solve: $\log_4(2 \times 4^{x-2} - 1) + 4 = 2x$

Q11 If $p = \log_a bc$, $q = \log_b ca$, $r = \log_c ab$, then prove that $pqr = p + q + r + 2$.

Q12 Find the value of the expression $\frac{2}{\log_4(2000)^6} + \frac{3}{\log_5(2000)^6}$

Q13 Simplify

$$\frac{81^{\frac{1}{\log_5 9}} + 3^{\frac{3}{\log_{\sqrt{6}} 3}}}{409} \cdot \left((\sqrt{7})^{\frac{2}{\log_{25} 7}} - (125)^{\log_{25} 6} \right)$$



Answer Key

Q1 a) 140

b) 12

c) 47

Q2 $x = 23 \text{ or } \frac{-9}{5}$

Q3 $x = 2 - \sqrt{2}$

Q4 $x = 100$

Q5 (B)

Q6 (A)

Q7 4

Q8 23

Q9 $x = \frac{7 \log(8) - 8 \log(7)}{\log(7) - \log(8)}$

Q10 $x = 2$

Q11 Proof

Q12 $\frac{1}{6}$

Q13 1



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